



**PROJECT REPORT
OF INDUSTRY ORIENTED HANDS ON EXPERIENCE
(IOHE)**

On

RAPZIS- Road Accident Prone Zone Identification System

COMPUTER SCIENCE AND ENGINEERING

B.E. Batch- 2022



**Under the guidance of:
Dr. Monika Aggarwal**

**Submitted By:
Mehak Garg - 2210990572**

**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
CHITKARA UNIVERSITY
PUNJAB**

ACKNOWLEDGEMENT

It is my pleasure to be indebted to various people, who directly or indirectly contributed in the development of this work and who influenced my thinking, behaviour and acts during the course of study.

I express my sincere gratitude to all for providing me an opportunity to undergo Industry Oriented Hands on Experience (IOHE) as the part of the curriculum.

I am thankful to “Dr. Monika Aggarwal” for her constant inspiration, presence, blessings, and for her invaluable support, cooperation, motivation, valuable suggestions, and precious time during the training and in accomplishing our project report.

Lastly, I would like to thank the almighty and our parents for their moral support and friends with whom I shared my day-to day experience and received lots of suggestions that improve our quality of work.

Name: Mehak Garg
Roll No.: 2210990572

1. ABSTRACT

Road accidents have become one of the leading causes of fatalities and injuries worldwide posing a serious threat to public safety and economic stability. The increasing number of vehicles, poor road conditions, violation of traffic rules and lack of awareness among drivers significantly contribute to rising accident rates. Traditional systems mainly focus on recording accident data but lack the capability to analyse patterns or predict future risks effectively. This creates a strong need for intelligent, data-driven systems that can assist in decision making.

This project, titled **RAPZIS (Road Accident Prone Zone Identification System)**, aims to provide an intelligent solution for analyzing road accident data and predicting accident risks using machine learning and GIS Mapping techniques together. The system integrates a MERN stack web application with a machine learning model developed in Python. The model is trained using historical accident data and predict risk levels as Low and High. In addition, clustering techniques such as K-Means and DBSCAN are applied to identify accident prone zones and detect dense clusters of accident occurrences.

The system provides a user-friendly interface where users can submit accident reports which are processed in real time generate predictions. It also includes an interactive dashboard with dynamic chart and graphs that visualize accident trends, severity distribution and environmental impacts such as weather conditions.

Overall, RAPZIS serves as a comprehensive platform that combines web technologies, machine learning and GIS Mapping to support authorities, urban planners and individuals in making informed decisions. The system has the potential to significantly contribute to improving road safety, reducing accident rates and promoting awareness through data driven analysis and predictive intelligence.

2. INTRODUCTION

Road accidents are a major global issue that leads to significant loss of human life, serious injuries and substantial economic damage each year. Various studies indicate that a large proportion of these accidents occur due to human factors such as overspeeding, distracted driving, driving under the influence of alcohol, violation of traffic rules and lack of awareness regarding road safety measures. Environment conditions like poor weather, low visibility and improper road infrastructure further contribute to the severity and occurrence of accidents.

The RAPZIS (Road Accident Prone Zone Identification System) project is designed to address these challenges by developing an intelligent platform that combines data analytics, machine learning and interactive visualization. By leveraging historical accident datasets, the machine learning model is capable of identifying patterns and predicting whether a situation falls under low or high risk. In addition to prediction, the system employs clustering techniques such as K-Means and DBSCAN to identify accident prone zones, which can help authorities take preventive measures in high risk areas.

Another key aspect of RAPZIS is its user-friendly and interactive web interface built using the MERN stack. The frontend provides intuitive forms for data entry, while the backend ensures efficient data handling and communication with the machine learning model. The system also includes dynamic dashboards and visualizations allowing users to easily understand accident trends and patterns.

Overall, the RAPZIS system aims to transform the way accident data is utilized by shifting from passive data recording to active prediction and analysis. It serves as a valuable tool for government authorities, traffic management departments and even individual users by providing insights that can contribute to improved road safety, better planning and reduces accident rates in the future.

3. METHODOLOGY

The development of RAPZIS – Road Accident Prone Zone Identification System follows a structured methodology that includes data processing, model building and system integration.

The methodology is divided into the following key phases:

3.1 Data Collection

The road accident dataset used in this project is collected from reliable and authentic sources such as Kaggle to ensure accuracy and consistency. The dataset contains various important attributes such as overspeeding, lane jumping, overtaking, wrong-side driving, alcohol consumption and other contributing factors. These features are essential for understanding the root causes of accidents and identifying patterns across different regions and time periods.

3.2 Data Preprocessing

Data preprocessing is performed to clean and prepare the dataset for analysis. Missing and null values are handled by either removing or replacing them to maintain data integrity. Unnecessary or redundant columns are dropped to reduce complexity. Column names are standardized for consistency and categorical variables such as state and region are converted into numerical format using encoding techniques.

3.3 Feature Engineering

Feature engineering is carried out to enhance the quality of input data. New features such as total accident counts are created by combining multiple attributes. Relevant features that significantly impact accident risk are selected while less important ones are excluded. The dataset is also transformed into a classification problem by defining a target variable that categorizes accident risk into levels such as Low and High. This step helps the model learn meaningful relationships between input features and the predicted outcome.

3.4 Clustering Techniques

Clustering techniques are applied to identify accident prone zone and patterns in the dataset. K-Means is used to group data points into clusters based on similarity. DBSCAN is also applied to detect dense regions and noise points which is useful in identifying high risk zones where accidents occur frequently. These clustering methods provide valuable insights into geographical and behavioural patterns associated with accidents.

3.5 Machine Learning Model Development

The dataset is divided into training and testing tests to evaluate model performance effectively. The model is trained on historical accident data, allowing it to learn patterns and relationships between different features. Multiple machine learning algorithms are implemented to predict accident risk levels such as Logistic Regression, Decision Tree and Random Forest. Among all, the Random Forest demonstrates the best performance due to its ability to handle complex data and reduce overfitting.

3.6 Model Deployment (Flask API)

After training and evaluation, the final model is saved as a serialized file (model.pkl) using pickle library. A Flask-based API is developed to integrate the machine learning model with the web application. The API receives input data from the frontend, processes it and returns the predicted risk level. This allows real time interaction between the web application and the machine learning model enabling users to obtain instant predictions.

3.7 MERN Stack Integration

The system is integrated using the MERN stack which includes MongoDB, Express.js, React.js and Node.js. The React.js frontend provides a user-friendly interface for data input, visualization and interaction. The Node.js and Express.js backend handle API requests, manage application logic and communicate with the Flask API. MongoDB is used as the database to store user information and accident reports. This integration ensures a scalable, efficient and responsive system architecture.

4. TOOLS & TECHNOLOGIES

To ensure robust functionality, modern design, and scalability, the RAPZIS web application is developed using the MERN stack along with various supporting tools and libraries.

Below is a breakdown of the tools and technologies employed across different layers of the project:

Frontend Technologies:

- React.js: A JavaScript library used to build a fast, interactive, and responsive user interface using reusable components.
- CSS: Used to design a modern, mobile-friendly, and responsive layout.
- JavaScript (ES6+): For implementing logic, event handling, and interaction on the client-side.

Backend Technologies:

- Node.js: A runtime environment used to run JavaScript code outside the browser, primarily for the server-side logic.
- Express.js: A lightweight web application framework used to build RESTful APIs and handle backend routing.

Database

- MongoDB: A NoSQL database used for flexible and scalable storage of user profiles, roommate listings, preferences, chat messages, and expenses.

Machine Learning

- Python: A versatile language used for building machine learning models.
- Pandas and NumPy: Python libraries used for data manipulation and analysis.
- Scikit-learn: A python library that provides simple and efficient tools for building and evaluating Machine learning models.

Development & Version Control Tools

- Visual Studio Code (VS Code): The primary code editor used for writing and debugging code.
- Flask: A lightweight Python web framework used to create APIs..
- Jupyter Notebook: An interactive environment used for training machine learning models.

6. MAJOR FINDINGS AND RESULTS

Model Evaluation:

The performance of the machine learning model was evaluated using standard classification metrics. The results indicate that the model performs effectively in predicting accident risk levels.

- **Accuracy:** 81% approx.
- **Precision:** 78% approx.
- **Recall:** 80% approx.
- **F1 Score:** 80% approx.

These metrics demonstrate that the model maintains a good balance between correctly predicting accident risks and minimizing errors.

Model Performace:

Different machine learning algorithms were implemented and compared to identify most suitable model for accident risk prediction.

- **Logistic Regression:** 65% accuracy.
- **Decision Tree:** 72% accuracy.
- **Random Forest:** 78% accuracy.

From the comparison, the Random Forest algorithm provides the highest accuracy and better generalization making it the most suitable model for this system.

7. CONCLUSION AND FUTURE SCOPE

Conclusion:

The RAPZIS system successfully demonstrates the effective integration of machine learning with modern web technologies to address the critical issue of road accidents. By combining data analysis, predictive modelling and interactive visualization, the system provides accurate risk prediction and meaningful insights based on user inputs. The use of the Random Forest algorithm ensures reliable performance, while clustering techniques help in identifying accident prone zones. Additionally, the MERN stack based architecture enables a scalable, user-friendly and efficient platform for real time data processing and visualization. Overall, RAPZIS serves as a practical and intelligent solution that can assist authorities, planners and individuals in making informed decisions, ultimately contributing to improved road safety and reduced accident rates.

Future Scope:

- Integration with live traffic APIs for real time updates.
- GPS based tracking to identify accident prone locations dynamically.
- Development of a mobile application for wider accessibility.
- Implementation of deep learning models for improved prediction accuracy.
- Integration with government systems for large scale deployment.
- Real time accident heatmaps for better visualization of risk zones.

8. REFERENCES

- Scikit-learn Official Documentation
- React.js Documentation
- MongoDB Documentation
- Kaggle Accident Dataset